

Automated action and motion recognition for excavator operations using egocentric videos

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ARTICLE INFO

Keywords:

Excavator
Action recognition
Motion recognition
Egocentric video
Computer vision
Long short-term memory
Optical flow

ABSTRACT

This paper proposes an automated model for recognizing actions and their execution patterns (motions) in excavator earthmoving operations using egocentric videos, addressing limitations of third-person video-based approaches. The model comprises two modules. An action recognition module detects the bucket states via computer vision, and classifies the corresponding actions (digging, hauling, dumping, and moving) through rule-based logic. A motion recognition module recognizes bucket movement motions (vertical and reach) and swinging motions by analyzing temporal visual changes of the bucket and arm using Long Short-Term Memory (LSTM) and of the background using optical flow. Through experimental validation, the proposed model demonstrated high, stable, and consistent recognition performance, achieving F1 scores of 0.928 for action recognition, and 0.920, 0.891, 0.908 for bucket vertical movement, bucket reach movement, and swinging motions, respectively. These results indicate potential for practical applications including data-driven productivity reporting, motion-based operational feedback, and future integration into autonomous equipment systems.

1. Introduction

The construction industry plays a pivotal role in shaping built environments that respond to socio-economic needs while serving as a foundation of national economic growth [1,2]. However, this role continues to be constrained by the industry's chronically stagnant productivity whose effects have become increasingly visible—construction productivity has grown at a compound annual growth rate (CAGR) of just 0.4 %, compared to 2.0 % for the global economy and 3.0 % for manufacturing over the past two decades [3]. That is, the construction industry is now facing a crisis due to its low supply capacity resulting from stagnant productivity, thus failing to meet the increasing demand driven by urbanization and infrastructure expansion. Also, it is challenged with a projected shortfall of 40 trillion dollars in unmet demand by 2040, representing 20 % of the total expected demand in the sector [3]. Meanwhile, in advanced economies, low productivity coupled with high costs, inflation, and rising interest rates poses more pressing challenges because it squeezes already thin profit margins for construction businesses, contributing to a contraction in the construction market. Consequently, maintaining productivity at the planned level and further

enhancing it remains one of the most critical imperatives in the construction industry.

Productivity is generally defined as the ratio of output to input. In the construction industry, this concept is interpreted and applied in various ways depending on the measurement's context and purpose. Since construction projects comprise multiple trades with different output units, the cost performance index (CPI) and schedule performance index (SPI) are most widely adopted as representative indicators of project-level productivity, enabling integrated management of cost and schedule performance [4–6]. These project-centric approaches offer valuable insights into strategic decision-making and enables comprehensive benchmarking across different construction projects and organizations [7–11]. However, it often overlooks the detailed inefficiencies at the operational level, hindering the ability to identify the root causes of productivity decline. For example, this integrated index may mask the specific causes of delays, making it more difficult to determine whether the delays in workflow stem from equipment waiting time, unnecessary repetition of work, or failure of collaboration between workers. That is, project-level management can view the macro workflow but often lacks the necessary details for an accurate diagnosis and correction of

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<https://doi.org/10.1016/j.autcon.2025.106662>

Received 7 August 2025; Received in revised form 5 November 2025; Accepted 6 November 2025

Available online 10 November 2025

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inefficiencies arising at the action level.

Accordingly, task-level productivity analysis, which captures observable on-site operations, is required to complement project-level management and reveal detailed sources of inefficiency [12]. Task-level productivity analysis is generally performed by focusing on either workers or construction equipment, the primary agents of on-site operations [13,14]. Among these, equipment-centered analysis plays a particularly crucial role because the operation of construction equipment directly determines the pace and efficiency of earthmoving, material handling, and other core construction processes. In this context, vision-based action recognition provides a powerful means of conducting equipment-centered productivity analysis, as each task in construction is inherently composed of a repetitive sequence of equipment actions, which are readily quantifiable and can serve as objective, real-time indicators of task-level productivity [15–17]. Therefore, accurate action recognition enables real-time identification of productivity bottlenecks—such as unnecessary equipment movement, excessive idle time, and non-optimal work sequences—and provides immediate feedback to maintain planned productivity levels on construction sites [14].

Beyond productivity analysis, vision-based action recognition has recently gained attention as a foundational technology for productivity enhancement through construction automation, particularly in autonomous and remote-controlled equipment operations [18,19]. In this context, it is essential not only to recognize what action is performed but also how it is executed (i.e., motions in this study) to assess whether construction equipment operates as intended and to enable control and feedback for optimized performance. Among various camera viewpoints, egocentric videos are essential as fundamental data for autonomous equipment operations, as they provide a first-person perspective that closely reflects the operator's cognitive perception, thereby enhancing the accuracy of action and motion recognition and improving the precision of assistance systems and autonomous control algorithms [20–22].

To lay the groundwork for advanced task-level productivity analysis and the development of autonomous systems, this study aims to establish a framework for developing automated models capable of efficiently and effectively recognizing both actions and motions of construction equipment based on egocentric videos. This framework was focusing on excavators, the key machinery most frequently used on construction sites and required to carry out complex operations across multiple construction phases [23,24]. In this study, an action is defined as the fundamental unit of purposeful behavior that constitutes a task, representing a distinct operational state of construction equipment (e.g., digging, hauling, dumping, or moving). A motion refers to the specific physical pattern in which an action is executed, for instance, the bucket's vertical, reach, or swinging movements during excavation.

2. Literature review

Prior studies on construction equipment action and motion recognition were comprehensively reviewed to establish the foundation for developing an automated framework for recognizing excavator actions and motions from egocentric videos. This review analyzes the strengths and limitations of existing approaches to identify research gaps and inform the proposed model in this study.

Nevertheless, very few studies on action recognition based on egocentric videos have been conducted [25–27]. Martin et al. [25] proposed a hybrid deep learning framework combining a Transformer encoder and long short-term memory (LSTM) network for automatic excavator action recognition and temporal localization from untrimmed egocentric videos. Kim et al. [26] developed a hybrid action recognition model that integrates kinematic data collected from gyroscopes and accelerometers with egocentric videos captured by cabin-mounted cameras, utilizing a combination of Convolutional Neural Networks (CNNs) and LSTM networks. However, these studies still demonstrated relatively limited recognition performance compared to third-person

video-based approaches, which have been the predominant focus of research in construction activity recognition. Although they utilized egocentric videos, their models were limited to classifying high-level actions and did not explicitly predict fine-grained motions of excavator components. Moreover, some relied on additional onboard sensors in combination with cameras, which increases system complexity and reduces the practicality of implementation in real construction environments. Therefore, this study primarily aims to enhance action recognition performance using only egocentric videos to a level comparable to that of third-person video-based approaches, and further to develop an automated model capable of recognizing both actions and fine-grained motions of excavator operations.

While this study focuses on egocentric video, the majority of computer vision research on construction equipment has been conducted using third-person videos captured from surveillance cameras or drones. In this research domain, the scope of studies has progressively expanded from fundamental object detection [28–30] to more advanced tasks including tracking [31–33], pose estimation [34–36], and action recognition [37–39]. However, achieving reliable vision-based understanding of construction equipment operations in such complex and dynamic environments remains challenging, particularly for action recognition tasks. Accordingly, various studies across many fields are being conducted to address this challenge [40,41].

Previous studies have investigated computer vision-based action recognition methods to enable real-time monitoring of equipment operation and action classification using third-person videos. Ghelmani and Hammad [42] proposed a single-stage framework that improves the you only watch once (YOWO) method for excavator activity recognition by integrating temporal gradient data through knowledge distillation. Jung et al. [43] and Zhang et al. [44] classified the actions of several construction equipment using a spatial-temporal deep learning model based on third-person videos collected from surveillance cameras. Shin et al. [45] proposed a three-dimensional (3D) Residual Neural Network classifier using transfer learning and an image preprocessing technique for object enlargement and sharpness enhancement to improve action recognition of distant construction equipment captured from a surveillance camera. Yang et al. [46] suggested a two-stage framework that converts equipment components into graphs and then enters them into a residual fusion graph convolution network to overcome the limitations of existing image-based equipment motion recognition due to the complex construction site environment and occlusion problems. Chen et al. [14] developed a multi-stage framework integrating faster region based CNN (R-CNN) detection, deep simple online and realtime tracking (SORT) tracking, and 3D residual network (ResNet) classification to automatically recognize detailed activities of multiple excavators and calculate their productivity. Yao et al. [47] recognized the joint positions of the cab, boom, arm, and bucket using computer vision based on the side images of an excavator captured using a monocular camera, and combined them with the Denavit-Hartenberg Kinetic model and physical constraints to estimate the 3D poses of the excavator.

In summary, multiple studies have indeed achieved a certain level of action recognition based on computer vision; however, the third-person video-based approach has inherent limitations due to the physical separation between cameras and equipment [48]. At construction sites with high variability and complexity, occlusions often occur when an object is located between the camera and the equipment, significantly reducing recognition performance, and the camera installation itself may physically interfere with construction work. In addition, long-distance camera shooting makes it difficult to detect detailed changes or minute movements of construction equipment. That is, the third-person video-based action recognition has some limitations in ensuring stable recognition performance and recognizing the detailed execution patterns of actions (i.e., motions in this study). In this context, egocentric videos can serve as a promising alternative to overcome the inherent limitations of third-person video-based approaches. Even when adopting an egocentric video, however, it remains essential to detect consistently observable

components—such as the bucket and arm—that can maintain visual stability amid the complex and dynamic conditions of construction sites, and to recognize equipment actions based on their motion patterns. Furthermore, advancing from action-level to motion-level recognition enables a more fine-grained understanding of the operational process, facilitating detailed analysis of equipment productivity in real-world construction environments.

3. Materials and methods

This study aimed to propose a framework for developing an automated model that could recognize action types and their execution patterns (i.e., motion) using egocentric videos captured by a cab-mounted camera during excavator operations. In essence, the generalizable framework enables the development of excavator-specific models with consistently high recognition accuracy across diverse site conditions. Among the various operations performed using different attachments, the earthmoving operation—one of the most fundamental and frequently performed using a bucket—was selected as the target. The

earthmoving operation consists of four sequential and cyclical actions: (i) digging; (ii) hauling; (iii) dumping; and (iv) moving. Each action is performed as a combination of motions driven by the operation of key excavator components (e.g., boom, arm, and cab): (i) bucket vertical movement; (ii) bucket reach movement; and (iii) swinging. Therefore, the model, encompassing two functional modules, is designed to recognize the actions that comprise the earthmoving operation and the motions that characterize how each action is performed, as detailed in Fig. 1. In the action recognition module, the excavator's actions are recognized by first using computer vision to detect various bucket states (e.g., empty, loaded, and dumping) frame by frame from the egocentric video, and then applying rule-based classification. In the motion recognition module, the excavator's motions are recognized through two sub-modules—bucket movement motion recognition and swinging motion recognition—by leveraging temporal changes in the arm, bucket, and background as input features, which are analyzed using an LSTM network and an optical flow model.

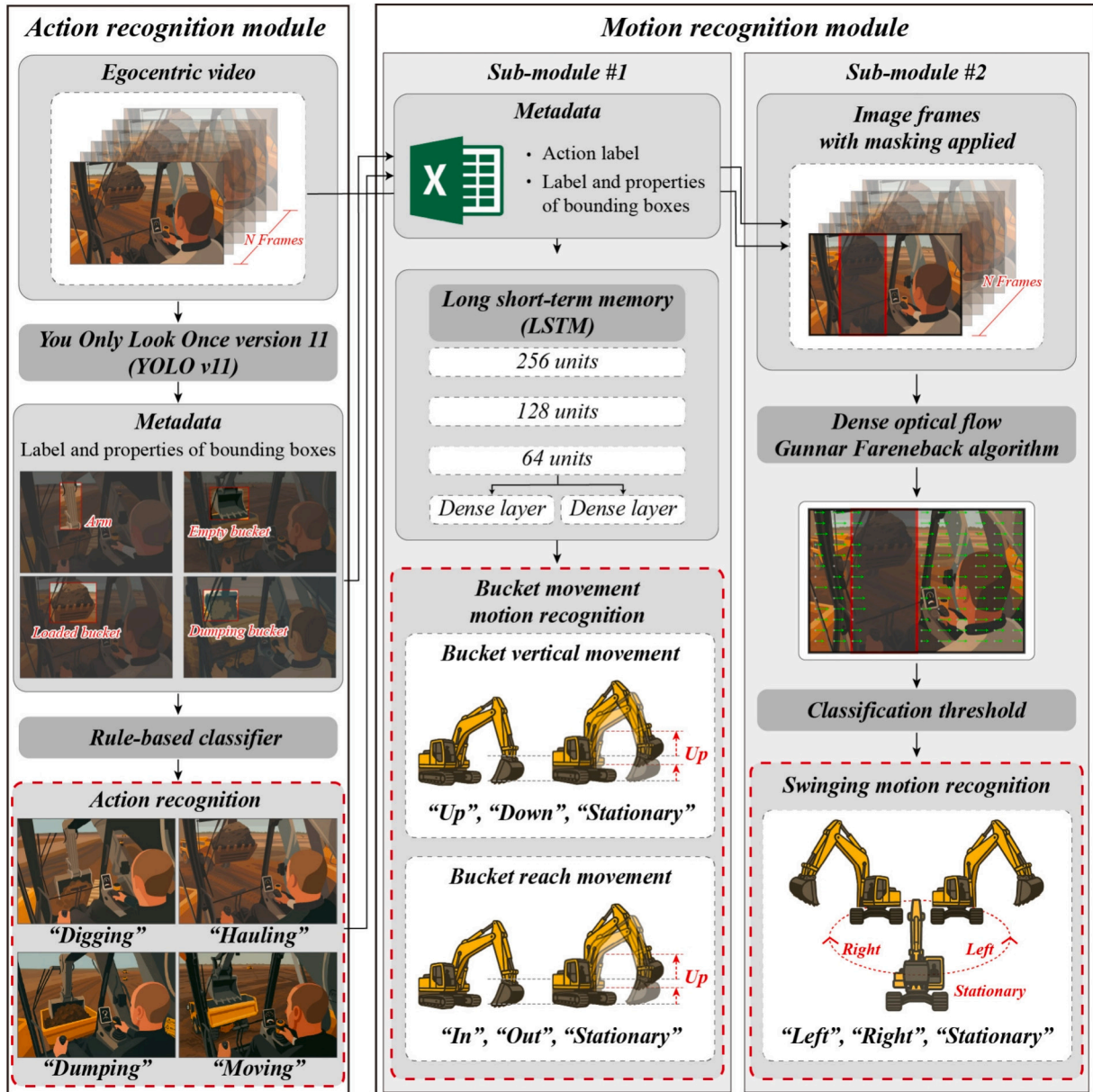


Fig. 1. Framework of the automated model for recognizing actions and motions in excavator earthmoving operations.

3.1. Action recognition module based on bucket state detection

The action recognition module, the first module of the proposed model, aims to recognize the four actions (i.e., digging, hauling, dumping, and moving) that comprise an excavator's earthmoving operation based on egocentric videos (refer to Fig. 2). In the egocentric video, components of the excavator—particularly the bucket—are consistently captured, allowing for detailed identification of their states as they change according to each action. Each action can be defined based on the bucket states, as follows. In digging, the empty bucket gradually disappears into the ground as it excavates soil. In hauling, the loaded bucket, filled with soil during digging, is transported to the dumping site. In dumping, the loaded bucket tilts to release the soil and gradually empties—a state defined as the dumping bucket, where the bucket and soil are visually observed together during dumping. In moving, the now-empty bucket travels to the next digging point. Consequently, the action recognition module, using computer vision techniques, can identify actions based on three bucket states (i.e., empty, loaded, and dumping) observed in the egocentric video.

First, a computer vision model is developed to detect the bucket in three distinct states from each frame of the egocentric video. To enable motion recognition in the subsequent module, arm detection is

incorporated, while the boom is excluded from detection due to its inconsistent appearance in the egocentric view. Then, considering the need for rapid inference speed for frame-by-frame detection and high precision for bucket states identification, You Only Look Once (YOLO) version 11 (YOLOv11) is selected as the computer vision model in this study [49]. YOLOv11 enhances real-time processing performance via a lightweight backbone network and improves accuracy in detecting differences in detailed visual information using advanced feature fusion techniques, while maintaining the rapid inference speed and high computational efficiency of the YOLO family of object detection models, which is considered a one-stage detector [50]. Subsequently, a training dataset for YOLOv11 is constructed based on annotated frames—each labeled with the empty bucket, loaded bucket, dumping bucket, and arm—extracted from two egocentric videos datasets prepared for different purposes (refer to Fig. 2 (a)). First, a general training dataset consisting of multiple egocentric videos (e.g., different camera installation conditions, excavator models, and site environments) not directly associated with the target excavator is incorporated to enhance training data diversity and improve the model's robustness. Second, an excavator-specific training dataset, consisting of an egocentric video of the target excavator covering five full cycles of the earthmoving operation, is incorporated to ensure data specificity for the detection task. In

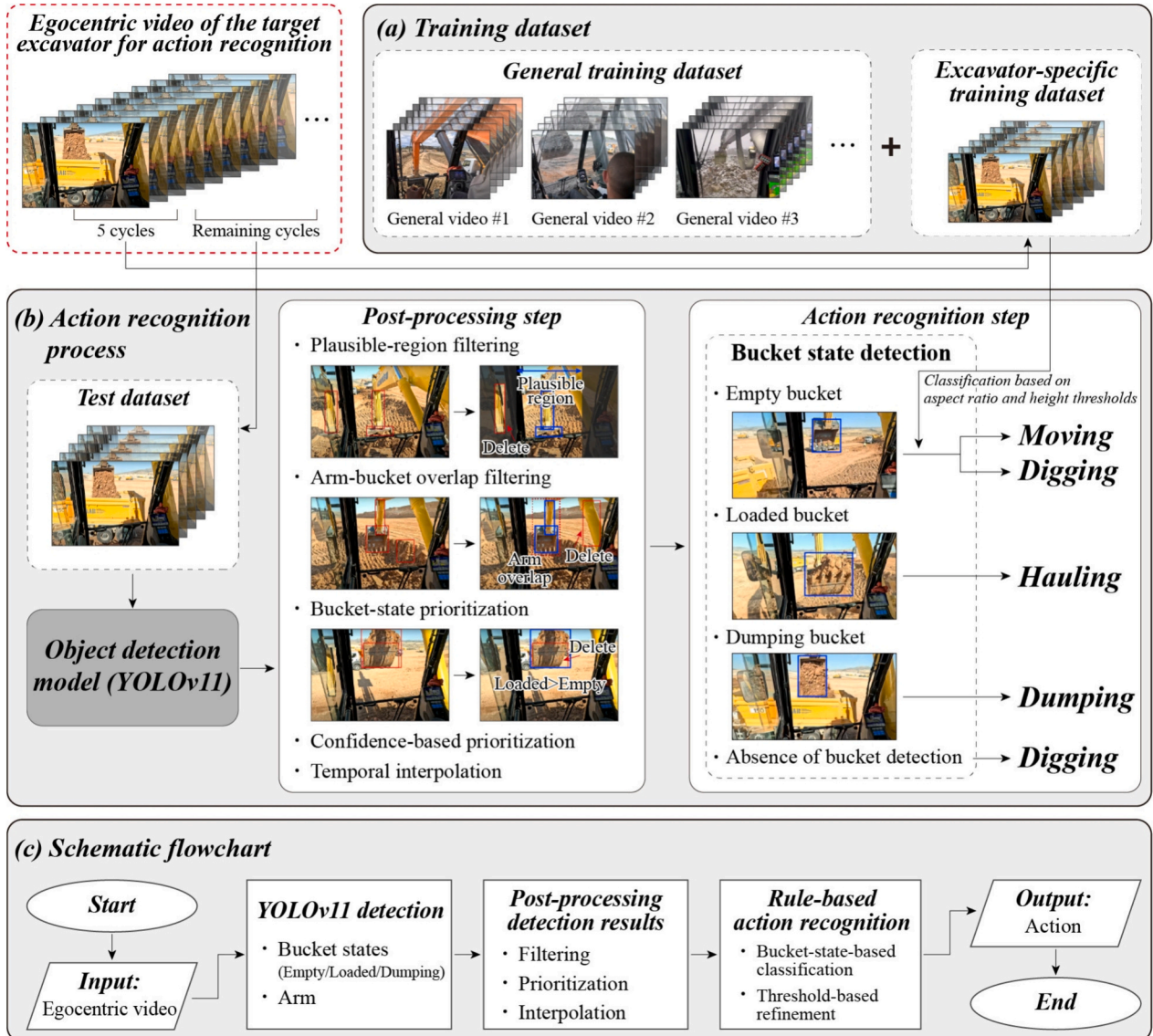


Fig. 2. Framework of the action recognition module based on bucket state detection.

other words, integrating these two datasets into a single training dataset ensures both stability and generalizability of the model: dataset specificity provides model performance stability under fixed conditions, while dataset diversity enhances the model's generalization ability and adaptability to various environmental changes in the field. To support subsequent recognition tasks, the excavator-specific training dataset is further annotated with frame-level action and motion labels in addition to the labels and properties (e.g., coordinates, width, height, area, and aspect ratio) of bounding boxes.

Second, post-processing is applied to increase the reliability of detection results, based on visual features observed throughout the earthmoving operation in the egocentric video (refer to the post-processing step in Fig. 2 (b)). The process comprises five sequential steps: (i) plausible-region filtering; (ii) arm–bucket overlap filtering; (iii) bucket-state prioritization; (iv) confidence-based prioritization; and (v) temporal interpolation. First, since components visible in the egocentric video (i.e., bucket and arm) move mainly within specific regions (i.e., plausible region) in the image, detections done outside these plausible regions were discarded. The plausible region is defined using the excavator-specific training dataset by excluding image areas where the bucket and arm were never detected, which correspond to the shaded areas in Fig. 2 (b). Specifically, while the bucket and arm can move freely in the vertical direction in egocentric videos, their horizontal movement is physically constrained within the excavator's operational range; therefore, the plausible region is established based on the horizontal bounds within which these components were consistently observed. This prevents false detections of components reflected in the cab's windshield or components from other excavators operating nearby. Second, since the arm is connected to the center of the bucket, any detected arm that did not overlap the x-coordinate range of the bucket, even if within the plausible region, is discarded. This prevents false detections of visually similar parts, such as the excavator's body frame or boom, from being misidentified as the arm. Third, since the bucket cannot exist in multiple states simultaneously, only one state is maintained with a redefined priority established in the following order: dumping, loaded, and empty. This priority reflects the logical hierarchy of bucket states. An empty bucket is less likely to be misdetected as a dumping or loaded bucket because it lacks content, including soil, whereas portions of a dumping or loaded bucket lacking soil may be misdetected as an empty bucket. In addition, while a loaded bucket is less likely to be misdetected as a dumping bucket because soil does not fall out, some portions of a dumping bucket, including soil, can be misdetected as loaded. Exceptionally, when an empty bucket is partially buried in the surrounding soil during digging, and a loaded bucket is also detected, the empty bucket label is prioritized if the spatial overlap between the bounding boxes is very high. This is because, when a loaded bucket and an empty bucket are detected simultaneously, the spatial overlap between the two bounding boxes of the two objects is relatively low due to soil contained in the loaded bucket. Fourth, since the excavator has one bucket and one arm, only the detections with the highest confidence scores were maintained for each. Fifth, since buckets and arms do not appear or disappear abruptly, interpolation is based on the preceding and following frames when there was a discrepancy of 0.1 s occurred in the detection results between consecutive frames. For example, if an empty bucket is continuously detected and then disappeared for a short period (e.g., less than 0.1 s), the empty bucket is considered present in the frame, and the coordinates of the bounding box were linearly interpolated based on adjacent frames. Conversely, if the empty bucket is not continuously detected and then detected only for a short time, it is considered not to be detected in the frame. This prevents false negatives and false positives caused by limitations in the developed computer vision model's performance. Consequently, after post-processing, metadata is extracted for all frames in the egocentric videos, including the labels and properties of detected bounding boxes.

Third, actions are recognized using a rule-based approach based on the metadata of detection results (refer to the action recognition step in

Fig. 2 (b)). A sliding window approach with overlapping steps is employed, as actions represent temporal processes or state changes that require analysis across multiple frames [51]. The use of overlapping steps mitigates discontinuities in predictions at action boundaries by promoting smooth transitions between actions. The window size is determined to ensure that no more than two distinct actions are included within a single window. The first step involves classifying actions based on bucket states for each frame within a window as follows: (i) empty bucket: digging or moving; (ii) loaded bucket: hauling; (iii) dumping bucket: dumping; and (iv) absence of bucket detection: digging. If a single action type, determined by the remaining bucket states excluding the ambiguous case of empty buckets, exceeds 50 % of the number of frames in the window, that action is set as the final action of the window. Otherwise, the characteristics of the empty bucket are analyzed to distinguish between digging and moving to determine the action. If the empty bucket gradually disappears into the ground during the digging process, the width remains relatively constant, while the height gradually decreases. Therefore, digging is recognized when the change in aspect ratio within the window exceeds a predefined aspect ratio threshold or when the minimum height falls below a height threshold. All remaining cases are classified as moving. These thresholds are optimized through grid search on the excavator-specific training dataset by evaluating various threshold combinations and selecting those that maximize classification accuracy between digging and moving actions. As with the preceding post-processing, short-term inconsistencies in action recognition lasting less than one second within a sequence of windows identified as the same action are interpolated based on the actions of adjacent windows.

In summary, the action recognition module processes egocentric video through three main stages (refer to Fig. 2 (c)): (i) YOLOv11 detection to identify bucket states and the arm; (ii) post-processing to refine detection results via filtering, prioritization, and interpolation; and (iii) rule-based action recognition that combines bucket-state-based classification with threshold-based refinement to produce the final action outputs.

3.2. Motion recognition module from temporal changes in visual features

The motion recognition module, the second module of the proposed model, aims to recognize the three motions (i.e., bucket vertical movement, bucket reach movement, and swinging) that comprise each action. Motion recognition plays a key role in analyzing the patterns of how each action is performed. In the egocentric video, temporal changes in visual features (e.g., bucket, arm, and background) occur whenever the excavator performs motions associated with each action, as driven by the operation of its components. Since the cab-mounted camera continuously captures the scene from a fixed viewpoint on the cab, the types of changing visual features vary depending on whether motions occur independently of the cab (i.e., bucket movement motion), or in conjunction with the cab (i.e., swinging motion). In the former case, the bucket, arm, and boom operate independently of the camera, resulting in dynamic visual changes in the bucket and arm (i.e., component dynamics) in the egocentric video. In the latter situation, the cab's operation causes the camera to move, leading to dynamic visual changes of the background (i.e., background dynamics) with the bucket and arm relatively stationary in the egocentric video. Consequently, the motion recognition module incorporates two distinct sub-modules to recognize each motion present in the egocentric video: component dynamics analysis using an LSTM-based model and background dynamics analysis using an optical flow model. When all motions are recognized as stationary, the excavator can be considered idling during that specific action.

3.2.1. Bucket movement motion recognition from component dynamics

The first sub-module (i.e., bucket movement motion recognition module) aims to recognize bucket movement motions, including vertical

movements (e.g., up, down, and stationary) and reach movements (e.g., in, out, and stationary) of the bucket, determined by the operation of the bucket, arm, and boom, based on component dynamics in egocentric videos (refer to Sub-module #1 in Fig. 1). Excavating operations are centered around the bucket. Although the boom and arm often move together in predictable patterns, these movements do not always translate directly to the bucket's motion. Depending on the coordination of components, the bucket may move differently, reflecting more complex interactions. In earthmoving operations, each action follows a consistent bucket movement sequence. For example, during hauling from the ground to a dump truck, the bucket typically follows an upward trajectory. In contrast, during the digging process, the bucket extends outward and then retracts inward, following a motion pattern that collects soil. Meanwhile, since this motion pattern is limited by the excavator's operating range, it is possible to maintain a single continuous motion. Due to the bucket's limited upward range, the bucket's motion must transition to a downward or stationary state after reaching its maximum height, causing each motion to form a sequential pattern. Therefore, in this study, bucket movement motions are defined as a series of bucket-based sequences, and a sequence model is applied to predict them effectively.

In recent studies on construction activity recognition, a variety of sequence modeling approaches, including Temporal Convolutional Networks (TCN) and Transformers, have been employed [35,42,44,52]. Among these alternatives, an LSTM-based model was selected for bucket movement recognition in this study, considering both the nature of motion recognition and practical implementation requirements. LSTM is particularly well-suited for this task because motion recognition fundamentally requires distinguishing subtle temporal patterns. Unlike action recognition, which identifies discrete action types based on bucket states, motion recognition must capture continuous behavioral characteristics—such as distinguishing smooth continuous movement from interrupted motion with pauses or identifying gradual directional changes versus abrupt transitions. These nuanced patterns are critical motion-based productivity analysis, as they reveal how efficiently tasks are performed rather than merely what tasks are performed. LSTM's recurrent architecture with memory cells is specifically designed to model such sequential dependencies and temporal dynamics. Additionally, practical considerations support this choice [24,25,44]. First, given the limited size of motion-related datasets in construction domains, LSTM provides more stable training and requires fewer parameters compared to Transformer or TCN architectures, which typically demand larger datasets to achieve optimal performance. Second, LSTM enables frame-by-frame sequential inference without requiring full-sequence context, making it suitable for real-time applications. Finally, the relatively straightforward architecture of LSTM facilitates model interpretation and practical deployment in construction site environments.

The LSTM-based model adopts a three-layer architecture (i.e., 256, 128, and 64 units) designed to hierarchically learn and analyze temporal patterns from short- to long-term dependencies, enabling effective recognition of bucket movement motions. The output of the LSTM-based model is fed into two dense layers, each configured to predict the vertical and reach movements of the bucket using a softmax function. This multi-output structure enables a single model to independently predict movements in both directions, allowing for accurate simultaneous recognition of both motions. The LSTM-based model is trained using the excavator-specific training dataset built in the previous step. As inputs, it receives action labels along with the labels and properties of bounding boxes, which are all derived from the action recognition module. As outputs, it produces frame-level predictions of bucket movements. Here, the properties of bounding boxes represent the dynamics of excavator components captured in the egocentric video. Finally, a sliding window approach, under the same conditions as the action recognition model, is employed to recognize bucket movement motions based on the mode within each window.

3.2.2. Swinging motion recognition from background dynamics

The second sub-module (i.e., swinging motion recognition module) aims to recognize the swinging motion of the bucket—controlled by the rotation of the cab—based on the background dynamics present in the egocentric video (refer to Sub-module #2 in Fig. 1). Since the cab-mounted camera also moves as the cab rotates, background dynamics that change in a consistent direction, rather than the movement of the arm and bucket, are observed in the egocentric video. Unlike component dynamics, background dynamics, which encompass an unspecified number of objects and highly variable construction sites, require an approach that captures consistent patterns of change across the entire field of view, rather than individual feature extraction. To this end, background dynamics are analyzed using optical flow, which represents visual motion information that shows the direction and extent of pixel movement within an image as a vector field based on pixel brightness changes between consecutive image frames [53].

The Gunnar Farneback algorithm, a dense optical flow method, is applied to calculate motion vectors for each pixel to capture consistent movement patterns of the entire background rather than feature-based tracking [54]. This algorithm utilizes quadratic polynomial approximation to preserve continuous flow between pixels, facilitating motion pattern estimation in the broad background of construction sites, and it exhibits robust performance in texture-rich environments. Based on these characteristics, swinging motion is recognized through the following processes.

First, to eliminate bucket movements occurring independently of the swinging motion, the areas corresponding to the bucket and arm are removed from the background. To do this, each image frame is offset on the x-axis by approximately 1 % of the total image width, serving as a minimal buffer, based on the bucket and arm positions detected by the action recognition model. This minimal offset accounts for bounding boxes that may not fully cover the bucket and arm and minor changes near these components (e.g., falling soil) that could interfere with background dynamic analysis. After the x-axis offset and y-axis range masking are applied to the image frame, the modified frame is input into the model. This prevents the bucket and arm movements from interfering with the estimation of consistent background motion patterns.

Second, since background changes unrelated to swinging can occur due to the high variability of construction sites, optical flow below a filtering threshold is removed, and the average of the remaining flow vectors is used to determine swinging motion. The filtering threshold is determined from the excavator-specific training dataset to suppress noise in non-swinging cases while preserving true swinging motions. This not only removes noise from the image itself but also prevents false positives due to local changes in the background, such as the movement of other construction equipment.

Third, given that the background moves in the opposite direction when the excavator swings in the x-axis direction, swinging is identified based on the mean of the x-axis components of the overall optical flow. When the mean x-axis component exceeds a positive swinging classification threshold, it is classified as left swinging; when it falls below a negative swinging classification threshold, it is classified as right swinging; otherwise, it is recognized as stationary. These thresholds are optimized through grid search on the excavator-specific training dataset by evaluating multiple candidate values and selecting those that maximize classification accuracy among left swinging, right swinging, and stationary.

As a result, the optical flow model takes each image frame as input and outputs the swing motion on a frame-by-frame basis. Finally, the same sliding window approach is employed to recognize the swinging motion based on the mode within each window.

4. Experiment procedure

To evaluate the feasibility and applicability of the proposed automated model for action and motion recognition in excavator operations

using egocentric videos, experiments were conducted using three videos collected from YouTube. Specifically, the selected videos, filmed in different working and shooting conditions, facilitated the evaluation of the proposed model's robustness in various real-world situations (refer to Fig. 3). Although the videos were sourced from YouTube, they captured authentic excavator operations filmed at real construction sites, representing genuine field data rather than controlled or simulated environments. This approach is consistent with established practices in construction automation research, where publicly available site videos are frequently used to validate models under diverse operational and environmental conditions, providing a practical means to assess real-world generalizability [41,46,55]. Video #1 is an egocentric video recorded using a camera mounted inside the cab of a yellow Komatsu excavator. The camera's viewpoint included the body frame, windshield, and side mirror, as well as the bucket, arm, and boom visible on the screen. Around the target excavator, multiple excavators and dump trucks were operating simultaneously while it excavated bright yellowish soil along an edge slope with distinct height differences. Video #2 is an egocentric video recorded using a camera mounted outside the cab of a black Hidromek excavator. Due to the camera's viewpoint, only the bucket, arm, and boom were captured on the screen. Around the target excavator, one loader and two dump trucks were operating simultaneously while it excavated reddish-brown soil on flat ground. Video #3 is an egocentric video recorded using a camera mounted outside the cab of an orange Hitachi excavator. Because the camera was positioned very close to the excavator, only the bucket and arm were captured in the frame, and the objects appeared significantly larger compared to other videos. The excavator was excavating brown soil that contrasted sharply with the color of its bucket. All videos have image frames with a resolution of 1280×720 pixels at 30 frames per second (FPS), and each video represents an independent continuous sequence of

earthmoving operations.

First, excavator-specific training datasets for all videos were constructed at the frame level, covering five full cycles of the earthmoving operation (i.e., digging-hauling-dumping-moving). The excavator-specific training datasets consisted of the labels and properties (e.g., coordinates, width, height, area, and aspect ratio) of bounding boxes and frame-level action and motion labels. For Video #1, a training dataset was constructed with 2721 image frames, corresponding to about 91 s of the video. The dataset included 1521 instances of empty buckets, 612 instances of loaded buckets, 425 instances of dumping buckets, and 2256 instances of arms. Each image frame was labeled with action (digging: 867, hauling: 617, dumping: 425, and moving: 812) as well as two motion types: bucket vertical movement (up: 668, down: 984, and stationary: 1069) and bucket reach movement (in: 1386, out: 656, and stationary: 679). For Video #2, a training dataset was constructed with 2601 image frames, corresponding to about 87 s of the video. The dataset included 1270 instances of empty buckets, 706 instances of loaded buckets, 331 instances of dumping buckets, and 2317 instances of arms. Each image frame was labeled with action (digging: 910, hauling: 721, dumping: 331, and moving: 639) as well as two motion types: bucket vertical movement (up: 766, down: 1061, and stationary: 774) and bucket reach movement (in: 498, out: 578, and stationary: 1525). For Video #3, a training dataset was constructed with 2997 image frames, corresponding to about 100 s of the video. The dataset included 1459 instances of empty buckets, 1005 instances of loaded buckets, 410 instances of dumping buckets, and 2997 instances of arms. Each image frame was labeled with action (digging: 877, hauling: 1039, dumping: 419, and moving: 662) as well as two motion types: bucket vertical movement (up: 1157, down: 1071, and stationary: 769) and bucket reach movement (in: 1062, out: 598, and stationary: 1337).

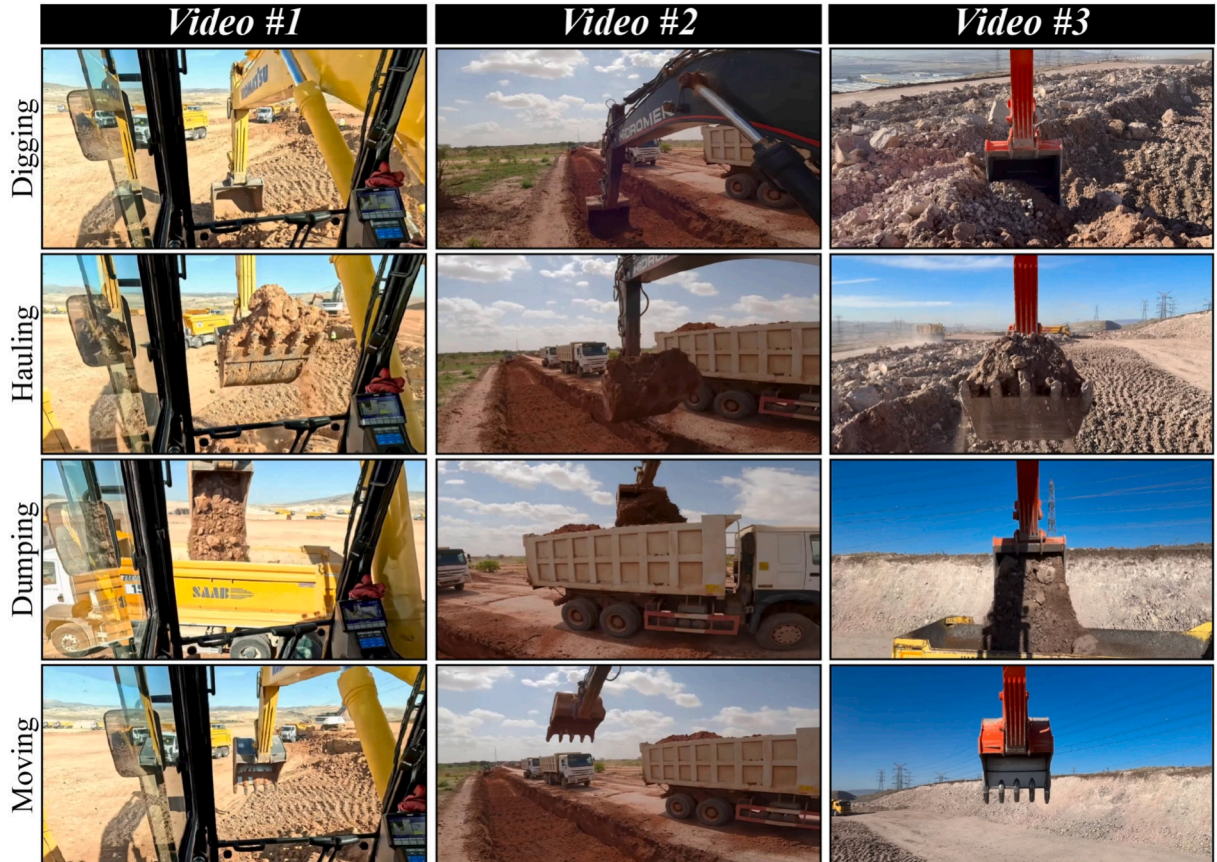


Fig. 3. Sample images of two egocentric videos.

Second, a YOLOv11-based computer vision model to detect empty bucket, loaded bucket, dumping bucket, and arm within the action recognition model was trained for each video. To minimize overlap between consecutive image frames, data were extracted at five-frame intervals from the excavator-specific training dataset. Also, to enhance dataset diversity, 1773 image frames (i.e., general training dataset) were extracted from 16 egocentric videos of earthmoving operations at different construction sites on YouTube, distinct from Videos #1 to #3. The training dataset was constructed by combining the general and excavator-specific training datasets from each video, as follows. For Video #1, 2319 annotated image frames, which included 1801 instances of empty buckets, 626 instances of loaded buckets, 270 instances of dumping buckets, and 2036 instances of arms, were used as the training dataset. For Video #2, 2294 annotated image frames, which included 1732 instances of empty buckets, 635 instances of loaded buckets, 260 instances of dumping buckets, and 2035 instances of arms, were used as the training dataset. For Video #3, 2373 annotated image frames, which included 1889 instances of empty buckets, 703 instances of loaded buckets, 275 instances of dumping buckets, and 2174 instances of arms, were used as the training dataset. All datasets were divided into training and validation sets at a ratio of 8:2. Then, to enhance the model's robustness and generalization performance, a two-stage augmentation process was applied. In the pre-training stage, pre-augmentation was performed using Roboflow with the following transformations: $\pm 15\%$ rotation, $\pm 10\%$ horizontal and vertical shear, grayscale conversion with a 15% probability, up to 2.5-pixel blur, exposure adjustment of $\pm 10\%$, and noise added to up to 0.1% of pixels [56]. In the training stage, on-the-fly augmentation was applied with the following transformations: $\pm 10\%$ translation, $\pm 50\%$ scaling, and horizontal flipping with a 50% probability. Additionally, in the hue-saturation-value (HSV) space, hue, saturation, and value were randomly adjusted by $\pm 1.5\%$, $\pm 70\%$, and $\pm 40\%$, respectively, while mosaic augmentation combined four image frames into one. The training process employed 100 epochs with a batch size of 16, implementing stochastic gradient descent optimization. Learning rate scheduling followed a cosine annealing approach, decreasing from an initial value of 0.01 to 0.0001, with an initial warm-up phase spanning the first 3 epochs.

Third, the LSTM-based model for recognizing bucket movement motions within the motion recognition module was trained using excavator-specific training datasets for each video. All datasets were split into training and validation sets with a ratio of 8:2. The training process employed 100 epochs and a batch size of 16 with Adam optimization. Learning rate scheduling followed an adaptive approach based on sparse categorical cross-entropy loss, starting at 0.1 and reduced by a factor of 0.5 when validation performance plateaued for 10 epochs. Early stopping and model checkpointing mechanisms were applied to ensure training stability. To prevent overfitting during training, dropout with a rate of 0.2 and batch normalization were applied.

Fourth, test datasets for all videos were constructed at the window level, covering a total of 15 full cycles of the earthmoving operation that were excluded from the excavator-specific training datasets. In other words, the test datasets were completely separated from the training datasets, thereby preventing potential overfitting during the evaluation of recognition performance. The window and step sizes were defined as 20 frames and 10 frames, respectively. For Video #1, a test dataset was constructed with 844 sliding windows, comprising 8450 image frames and about 282 s of video. Each window was labeled with action (digging: 268, hauling: 204, dumping: 149, and moving: 223) as well as three motion types: bucket vertical movement (up: 211, down: 341, and stationary: 292), bucket reach movement (in: 425, out: 171, and stationary: 248), and swinging (left: 219, right: 175, and stationary: 446). For Video #2, a test dataset was constructed with 815 sliding windows, comprising 8160 image frames and about 272 s of video. Each window was labeled with action (digging: 259, hauling: 255, dumping: 110, and moving: 191) as well as three motion types: bucket vertical movement (up: 226, down: 319, and stationary: 270), bucket reach movement (in: 197, out:

161, and stationary: 461), and swinging (left: 156, right: 188, and stationary: 471). For Video #3, a test dataset was constructed with 891 sliding windows, comprising 8920 image frames and about 297 s of video. Each window was labeled with action (digging: 289, hauling: 280, dumping: 111, and moving: 211) as well as three motion types: bucket vertical movement (up: 352, down: 332, and stationary: 206), bucket reach movement (in: 280, out: 98, and stationary: 513), and swinging (left: 247, right: 290, and stationary: 354).

Fifth, the recognition performance of the developed model for action and motion recognition was evaluated based on classification metrics used in multi-class classification tasks. The recognition results were first categorized into four cases by comparing them with the ground truth, thereby forming a confusion matrix in a one-vs-rest manner for each class [57]: (i) true positive (TP); (ii) false positive (FP); (iii) false negative (FN); and (iv) true negative (TN). Based on this confusion matrix, the recognition performance was quantitatively assessed using three performance metrics (refer to Eqs. (1) to (3)): (i) precision; (ii) recall; and (iii) F1 score. Precision represents the level of confidence in positive results predicted by the model, while recall indicates how accurately the model recognizes actual positive cases [58–60]. The F1 score is the harmonic mean of precision and recall [61,62]. Among the three, the F1 score was used as a representative metric to evaluate recognition performance for individual classes as well as for all classes. In contrast, precision and recall were employed to analyze the specific characteristics of recognition performance in detail. These metrics have values between 0 and 1, with higher values indicating better recognition performance. The metrics for all classes were calculated using macro averaging.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

$$\text{F1 score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

Where *TP*, *FP*, *FN* stand for true positive, false positive, and false negative.

5. Experimental results

This section presents the experimental results of the proposed automated model. The performance of each module—action recognition and motion (e.g., bucket movement motion and swinging motion) recognition—is analyzed and discussed to evaluate the model's overall effectiveness.

5.1. Action recognition

Actions were recognized and classified into digging, hauling, dumping, and moving via rule-based classification for bucket states (i.e., empty, loaded, and dumping) detected based on the computer vision model, and action recognition performance was analyzed (refer to Table 1 and Fig. 4). The overall recognition performance across all videos was high and stable, gaining an F1 score of 0.928 with well-balanced precision (0.931) and recall (0.926) that differed by only 0.5%, indicating consistent action recognition. The F1 scores of individual action classes were relatively higher in “hauling” (0.964) and “dumping” (0.976) than in “digging” (0.894) and “moving” (0.879). These results suggest that while the reliable detection of bucket states enabled efficient recognition of “dumping” and “hauling” only by detecting a single object (i.e., loaded bucket and dumping bucket), the performance of recognition for “digging” and “moving” was relatively low due to the empty bucket-based classification criteria. The analyzed action recognition performance for each video is described as follows:

Table 1
Action recognition performance.

Video type	Action class	Performance metric		
		F1 score	Precision	Recall
All videos	All classes	0.928	0.931	0.926
	Digging	0.894	0.889	0.900
	Hauling	0.964	0.972	0.957
	Dumping	0.976	0.989	0.964
	Moving	0.879	0.873	0.884
Video #1	All classes	0.906	0.913	0.900
	Digging	0.877	0.852	0.903
	Hauling	0.916	0.952	0.882
	Dumping	0.951	0.986	0.919
	Moving	0.879	0.862	0.896
Video #2	All classes	0.937	0.939	0.936
	Digging	0.904	0.900	0.907
	Hauling	0.982	0.977	0.988
	Dumping	0.982	0.991	0.973
	Moving	0.881	0.888	0.874
Video #3	All classes	0.941	0.940	0.943
	Digging	0.902	0.915	0.889
	Hauling	0.993	0.986	1.000
	Dumping	0.996	0.991	1.000
	Moving	0.875	0.869	0.882

Overall, the action recognition showed high performance and

consistency across all videos. The overall F1 scores increased in the order of Video #1 (0.906), Video #2 (0.937), and Video #3 (0.941), with precision-recall differences remaining minimal, ranging from 0.3 % to 1.3 %, indicating well-balanced recognition stability. Furthermore, across all videos, higher F1 scores were consistently achieved for “dumping” and “hauling” than for “digging” and “moving”. Afterward, a confusion matrix and misclassified image frames were analyzed to identify the root causes of performance degradation. The analysis found that classification errors occurred mainly during transitions between consecutive actions, particularly during the transition from “moving” to “digging” and from “digging” to “hauling”. During the transition between “moving” and “digging”, classification is based on the aspect ratio change and the height of the empty bucket. However, if these criteria are not satisfied, classification errors occur more frequently than in other action transitions or sections. This is the main reason for the lowest recognition performance of “moving” and “digging”. Also, in the transition section between “digging” and “hauling”, the delay in detecting the loaded bucket due to soil loss during the soil lifting process caused misclassification from “hauling” to “digging”, leading to a slight reduction in the recall performance of “hauling” in Video #1.

Next, the action recognition performance of this study was compared with that of existing action recognition studies (refer to Fig. 5). As comparison targets, three recent studies that recognized digging, hauling, dumping, and moving were selected: two (i.e., Wang et al. [46] and

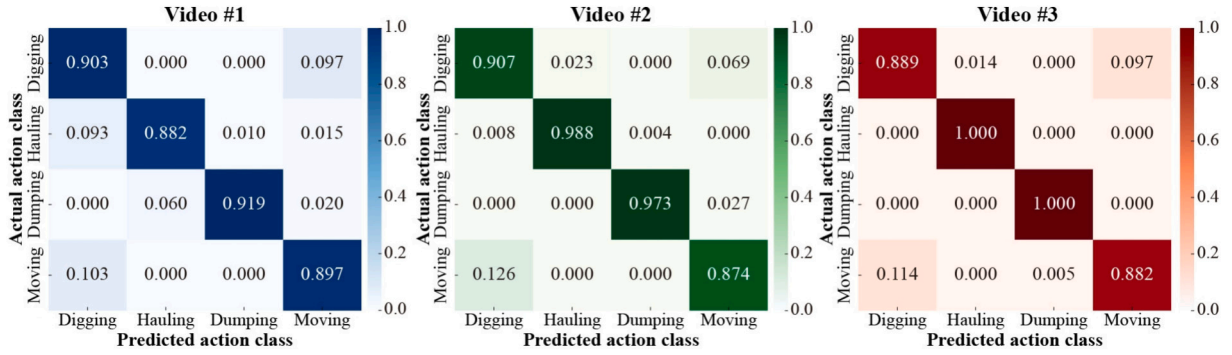


Fig. 4. Confusion matrices for action recognition in Videos #1, #2, and #3.

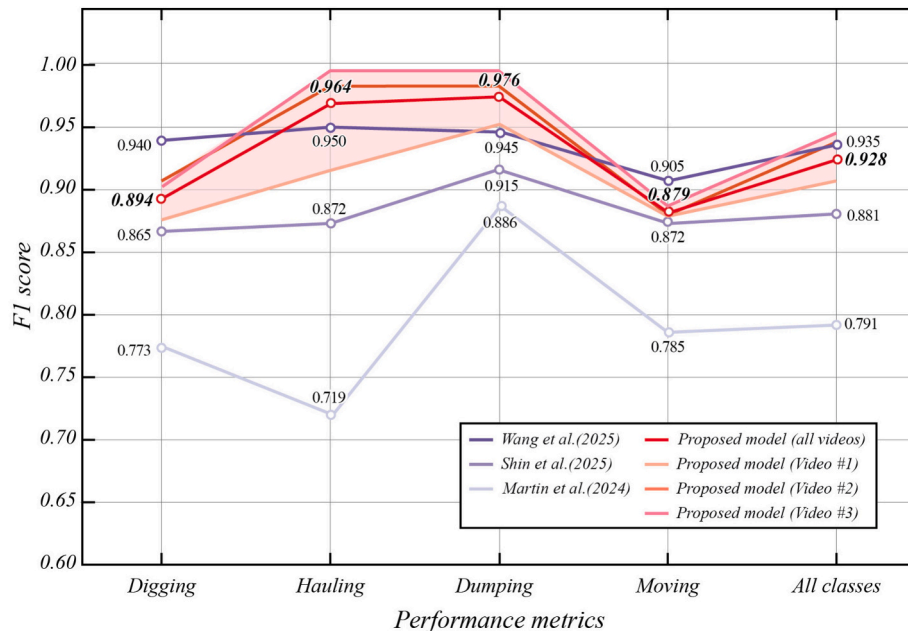


Fig. 5. Comparison of action recognition performance between the proposed model and previous studies.

Shin et al. [45]) based on third-person video and one (i.e., Martin et al. [25]) based on egocentric video. The proposed model demonstrated highly competitive action recognition performance across all classes, falling just 0.7 % short of the best-performing model (Wang et al.), while outperforming Shin et al. and Martin et al. by 4.7 % and 13.7 %, respectively. Notably, in Videos #2 and #3, the proposed model outperformed the best-performing model in action recognition across all classes. In the analysis of individual action classes, the proposed model outperformed the other two studies, except for Wang et al. Wang et al.'s research produced better results in "moving" and "digging", while the current study achieved a superior outcome in "dumping" and "hauling". Overall, this study demonstrated highly competitive action recognition performance results compared to previous studies. Then, given the performance limitations of third-person video-based research due to occlusion and other factors on actual construction sites, the egocentric-based action recognition approach of the present research is expected to have a greater practical value in real-world situations.

In summary, the action recognition module in this study demonstrated high and stable performance supported by the reliable detection of bucket states. This implies that the present research provides a simple yet robust mapping structure capable of achieving high action recognition performance solely via accurate bucket state detection in egocentric video. In addition, the proposed method has demonstrated competitive performance compared to existing third-person video-based action recognition methods and is expected to have a much greater practical value, considering the occlusion constraints of actual construction sites. It is also expected that the action recognition performance will be further improved, particularly by enhancing the distinction between "moving" and "digging," through the incorporation of additional visual context information capturing bucket-soil interactions or by adopting a context-aware machine learning-based classification approach.

5.2. Motion recognition

This section presents the experimental results of motion recognition, focusing on the recognition performance of bucket movements and swinging motions using the LSTM- and optical flow-based models.

5.2.1. Bucket movement motion recognition

The two bucket movement motion types were independently recognized using an LSTM-based model: (i) bucket vertical movement: up, down, and stationary; and (ii) bucket reach movement: in, out, and stationary. The recognition performance was analyzed for each motion (refer to Table 2 and Fig. 6). The overall recognition performance across all videos for vertical movements was high and stable, achieving an F1 score of 0.920 with well-balanced precision (0.926) and recall (0.916) that differed by only 1.0 %, indicating consistent recognition of this motion type. The F1 scores for each motion class were 0.938, 0.926, and 0.895, respectively, for down, stationary, and up, and the performance deviation among these classes was within 4.3 %. Compared to vertical movements, the recognition performance for reach movements was slightly lower but remained consistently high. The overall recognition performance across all videos for reach movements garnered an F1 score of 0.891 with precision (0.901) and recall (0.886) values showing only a 1.5 % difference. The F1 score was relatively low in classes "out" (0.820) compared to "stationary" (0.943) and "in" (0.912), with 12.3 % performance degradation relative to "stationary". This suggests that "out" was misrecognized as "in" or "stationary" in some cases. The recognition performance for vertical movement and reach movement in each video is explained as follows:

Overall, the bucket movement motion recognition demonstrated high performance; however, performance variations were observed depending on the type of bucket movement, as vertical and reach movements exhibit different visual patterns in egocentric videos. The vertical movement motion recognition showed high performance and

consistency in all videos, as the upward and downward movements of the buckets were clearly captured in egocentric videos, aligning with the vertical plane of the camera view. The overall F1 scores increased in order of Video #2 (0.900), Video #3 (0.926), and Video #1 (0.933), with precision-recall differences remaining minimal, ranging from 0.5 % to 2.0 %, indicating well-balanced recognition stability. However, only the F1 score of "up" was below 0.900, which was due to low recall. In contrast, the reach movement motion recognition demonstrated high performance but was relatively lower than vertical movement, as inward and outward movements occur along the depth axis perpendicular to the image plane, and egocentric videos lack explicit depth information to clearly capture these distance changes. The overall F1 scores increased in order of Video #2 (0.850), Video #3 (0.899), and Video #1 (0.926), and were consistently lower than those of vertical movement across all videos. Nevertheless, the precision-recall differences remained low, ranging from 0.3 % to 2.5 %. Furthermore, across all videos, the F1 score for "out" was the lowest, as the bucket and arm become progressively smaller as they move away from the camera, making it more difficult to detect changes.

A confusion matrix and misclassified image frames were analyzed to identify the causes of poor recognition performance for vertical and reach movement motions via the LSTM-based model (refer to Fig. 7). First, in certain situations, an insufficient number of bounding boxes detected failed to provide adequate input for the LSTM-based model, leading to a decrease in recognition performance. For example, in all videos, during the final stage of digging—when the soil is being lifted—the bucket was not detected, and only the arm was identified. As a result, although the actual state of the bucket was "up" in vertical movement and "stationary" in reach movement, the model incorrectly recognized the motions as "stationary" and "in", respectively. Likewise, during the hauling process in Video #2, the arm was not detected as the bucket was positioned above the egocentric view, causing classification errors for "up" and "out". Second, recognition performance declined as detection stability was compromised by significant fluctuations in bounding box coordinates across frames. For example, classification errors occurred in vertical movement and reach movement due to bounding box coordinate instability caused by soil loss during the transition from hauling to dumping, or bounding box instability of the empty bucket caused by soil movement during the digging process. Third, while the inclusion of the action label as an input variable generally improved the LSTM-based model's performance, issues related to action labels in certain cases adversely affected recognition performance. One such case involved misclassification by the action recognition module. For example, in Video #2, the transition from moving to digging was misclassified by as digging, causing the outward reach movement to be incorrectly classified as inward, since digging typically involves inward bucket motion for soil collection. Another case involved bias in the training data for specific actions. For example, in Video #3, the excavator-specific training dataset predominantly exhibited upward vertical movement immediately after dumping during moving, causing the model to misclassify stationary vertical movement observed at the same stage.

In summary, the LSTM-based bucket movement motion recognition module developed in this study demonstrated high and stable performance. That is, the bucket vertical movement and the bucket reach movement were successfully recognized by learning the visual change patterns of bounding boxes observed during earthmoving operations. Performance variations between the two motion types were observed, with vertical movement achieving higher accuracy than reach movement due to differences in visual representation—vertical movements align with the image plane while reach movements occur along the depth axis lacking explicit depth information. On the other hand, since the LSTM-based model relies on visual features extracted from the limited field of view provided by egocentric videos, a decline in recognition stems from the following factors: (i) restricted input data caused by missed detection of the bucket or arm; (ii) instability of bounding box

Table 2
Recognition performance for bucket movement motions.

Motion type	Video type	Motion class	Performance metric		
			F1 score	Precision	Recall
Bucket vertical movement	All videos	All classes	0.920	0.926	0.916
		Up	0.895	0.939	0.855
		Down	0.938	0.911	0.967
		Stationary	0.926	0.928	0.925
		All classes	0.933	0.945	0.925
		Up	0.895	0.972	0.829
	Video #1	Down	0.935	0.902	0.971
		Stationary	0.968	0.960	0.976
		All classes	0.900	0.903	0.898
		Up	0.869	0.904	0.836
		Down	0.932	0.920	0.944
		Stationary	0.900	0.885	0.915
	Video #2	All classes	0.926	0.930	0.924
		Up	0.920	0.941	0.901
		Down	0.948	0.911	0.988
		Stationary	0.910	0.938	0.883
		All classes	0.891	0.901	0.886
		In	0.912	0.893	0.933
Bucket reach movement	All videos	Out	0.820	0.857	0.789
		Stationary	0.943	0.951	0.936
	Video #1	All classes	0.926	0.940	0.915
		In	0.955	0.921	0.991
		Out	0.875	0.911	0.842
		Stationary	0.948	0.987	0.911
	Video #2	All classes	0.850	0.861	0.846
		In	0.844	0.800	0.893
		Out	0.758	0.841	0.689
		Stationary	0.948	0.942	0.954
	Video #3	All classes	0.899	0.901	0.898
		In	0.936	0.959	0.914
		Out	0.828	0.820	0.837
		Stationary	0.933	0.923	0.943

coordinates; and (iii) error propagation from action recognition module. Thus, the development of advanced computer vision-based action recognition modules, together with improved bucket and arm detection, is expected to overcome the limitations of the LSTM-based model. Moreover, integrating supplementary information, such as third-person videos or depth maps, can alleviate the inherent limitations of egocentric videos and enhance bucket movement motion recognition, especially for reach movements.

5.2.2. Swinging motion recognition

In swinging motion recognition, swinging motions were recognized as three classes (i.e., left, right, and stationary) based on the optical flow model, and the recognition performance was analyzed (refer to Table 3 and Fig. 8). The overall recognition performance across all videos was high and stable, achieving an F1 score of 0.908 with well-balanced precision (0.916) and recall (0.908) that differed by only 0.8 %, indicating consistent swing motion recognition. For the F1 score for each motion class, the highest values were 0.911 for “right”, followed by 0.909 for “stationary”, and 0.903 for “left”, respectively, and the performance deviation among these classes was within 0.8 %. These results suggest that the model has a balanced recognition capacity without being biased toward specific motion classes. The recognition performance for swinging motion in each video is detailed as follows:

Overall, the swinging motion recognition demonstrated high performance; however, noticeable performance variations occurred depending on whether the egocentric camera was mounted inside (i.e., Video #1) or outside (i.e., Videos #2 and #3) the cab. In Video #1, the camera mounted inside the cab introduced background distortions from the windshield reflections, leading to a relatively lower F1 score (0.831) and a larger precision–recall difference (5.3 %). A confusion matrix and misclassified image frames were analyzed to identify the causes of performance degradation (refer to Fig. 9). The lower performance in Video #1, primarily stemmed from the misclassification of “left” and “right” as “stationary” or the misclassification of “stationary” as “right”. The main reason for the misclassification of “left” and “right” as “stationary” was the offset motion vectors in the optical flow-based analysis.

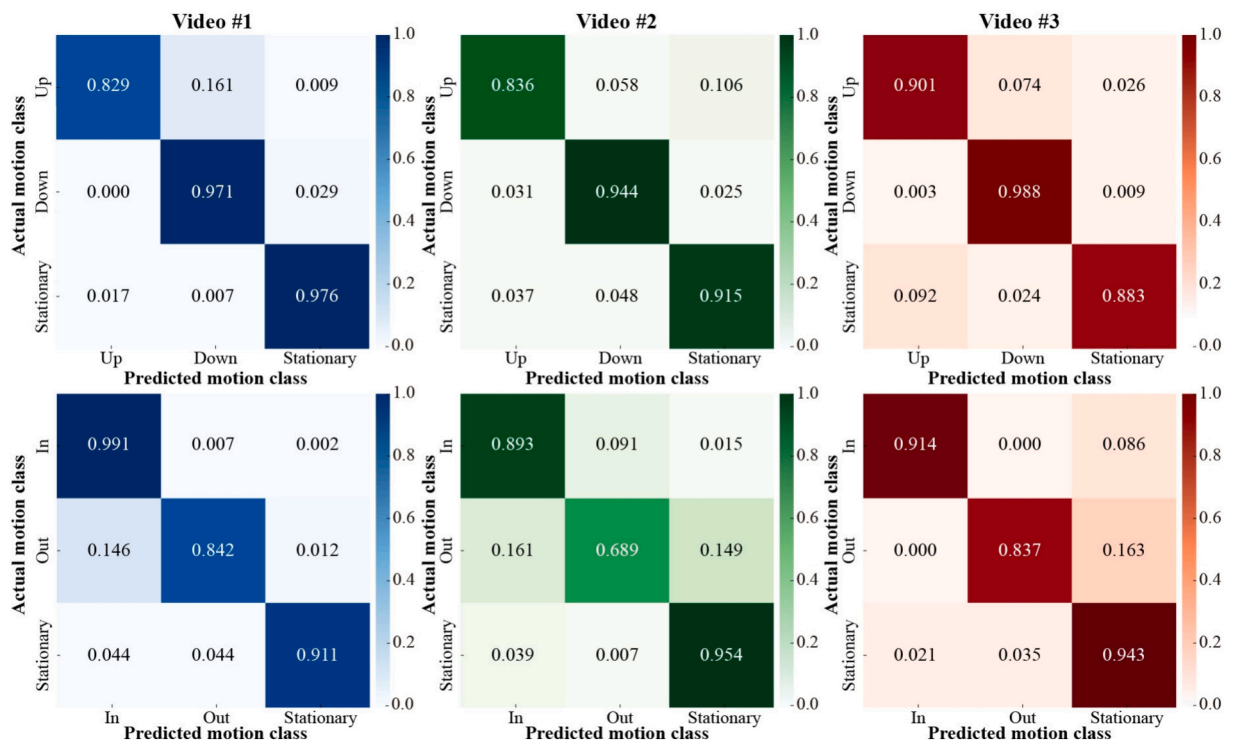


Fig. 6. Confusion matrices for bucket movement motion recognition in Videos #1, #2, and #3.

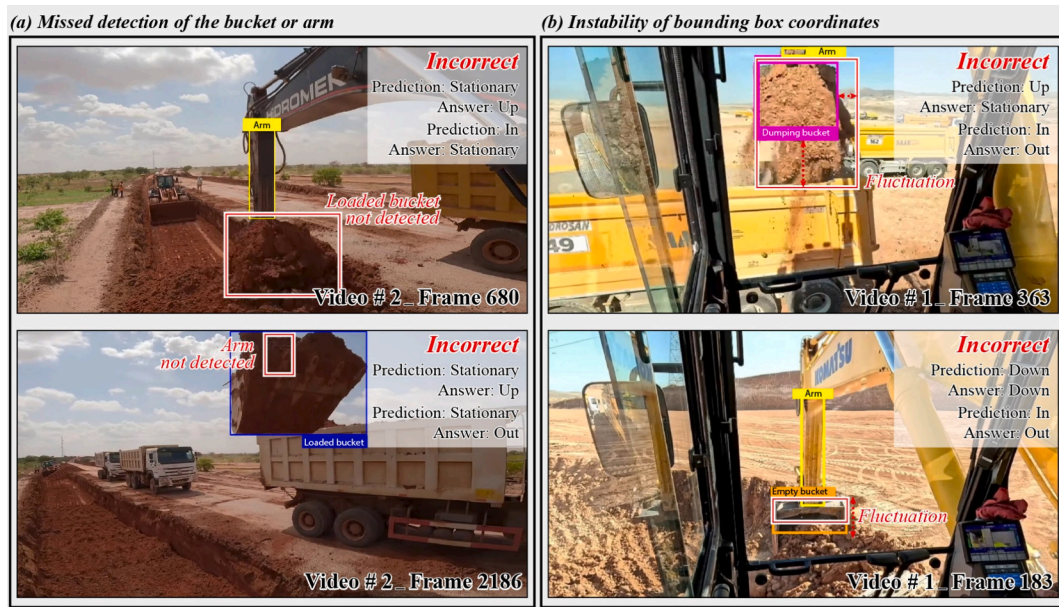


Fig. 7. Examples of incorrect bucket movement motion recognition.

In the video captured by the camera mounted inside the cab, the background and shadows reflected in the side mirrors and windshield shifted in different directions from the actual background, resulting in an offset in the overall motion. This led to the misclassification of “stationary” instead of oscillatory motion when the object was swinging to the “left” and “right”. Nevertheless, motion was accurately recognized in most frames despite reflections or shadows. Moreover, “stationary” was misclassified as “right” due to the rotation of the camera to the right with the cab fixed.

In contrast, Videos #2 and #3, where cameras were mounted outside the cab and captured the background without visual interference, achieved higher F1 scores of 0.970 and 0.922, respectively, with low precision-recall differences of 1.2 % to 1.6 %. While Video #2 demonstrated high recognition performance across all motion classes, Video #3 showed relatively lower precision (0.840) for “left” and lower recall (0.805) for “stationary”, primarily due to the misclassification of “stationary” as swing motions. The classification error observed in both Videos #2 and #3, though more pronounced in Video #3, was primarily due to low swinging speed at the start or end of the swinging motion.

In summary, the optical flow-based swinging motion recognition module developed in this study exhibited high and stable performance. However, performance differences due to the characteristics of egocentric videos, depending on the camera’s status and installation conditions, occurred. Such differences were linked to the need to process a large amount of visual information from a wider background area, other than action and other motions involving specific parts, such as the arm and bucket. Accordingly, to improve the performance of swinging motion recognition, proper camera installation is crucial and requires careful consideration for secure mounting and robust placement to prevent interference (e.g., reflections and shadows) from being captured in egocentric videos. Alternatively, introducing technology for identifying and processing noise in consistent background motion is expected to contribute to achieving higher recognition performance for swinging motion.

6. Discussion

This section discusses the key contributions, practical applications, and limitations of the proposed framework. The findings are interpreted in terms of their implications for construction productivity analysis and potential directions for future research and development.

6.1. Contributions and applications of the proposed model

The proposed framework uniquely integrates action and motion recognition from egocentric videos, providing a scalable foundation for near-future extensions into automated task-level productivity analysis and, ultimately, intelligent construction automation driven by autonomous equipment operation. Unlike previous approaches that focus solely on action-level analysis, this dual-level recognition capability allows both task-level productivity measurement (via actions) and fine-grained diagnosis (via motions), bridging the gap between what is being done and how efficiently it is being executed. The experiments demonstrated that the proposed model consistently achieved high performance in recognizing excavator actions and motions during earth-moving operations, confirming its robustness, generalizability, and adaptability to varying excavator types and operational environments. The proposed model has the following contributions:

The first contribution is establishing an egocentric vision-based framework that enables both action and motion recognition, offering strong potential for scalability across various applications beyond productivity analysis. In contrast, previous third-person approaches are constrained by occlusion and limited viewing angles, which hinder the reliable identification of excavator components and degrade action recognition performance. As a result, such methods remain confined to action-level recognition, providing only task-level productivity metrics without revealing how tasks are actually performed at the motion level. The proposed framework overcomes these limitations and further advances existing approaches by recognizing motions within actions to enable granular diagnosis of how tasks are performed, not just what tasks are performed. This distinction is critical for actionable productivity improvement, as demonstrated through motion-based feedback that provides interpretable insights for task optimization. Moreover, the motion-based feedback can be utilized not only as a guideline for workers but also as a valuable input for training and control in remote-operated and autonomous systems. Furthermore, more stable and precise recognition can be expected when egocentric and third-person view-based approaches are integrated in a complementary manner. Such integration could further improve the accuracy and efficiency of downstream tasks by reducing error propagation through the analytical pipeline.

The second contribution is the hybrid recognition architecture that combines computer vision-based action recognition with LSTM- and

Table 3
Recognition performance for swinging motions.

Video type	Motion class	Performance metric		
		F1 score	Precision	Recall
All videos	All classes	0.908	0.916	0.908
	Left	0.903	0.914	0.906
	Right	0.911	0.904	0.919
	Stationary	0.909	0.930	0.898
Video #1	All classes	0.831	0.864	0.811
	Left	0.829	0.958	0.731
	Right	0.802	0.829	0.777
	Stationary	0.861	0.805	0.926
Video #2	All classes	0.970	0.962	0.978
	Left	0.966	0.945	0.987
	Right	0.969	0.954	0.984
	Stationary	0.975	0.989	0.962
Video #3	All classes	0.922	0.922	0.934
	Left	0.913	0.840	1.000
	Right	0.962	0.929	0.997
	Stationary	0.891	0.997	0.805

optical flow-based motion recognition. The design is purposeful: actions (digging, hauling, dumping, and moving) are defined by discrete bucket states extracted from individual frames using a computer vision model, making rule-based classification both efficient and interpretable. In contrast, the motion recognition modules focus on modeling the continuous dynamics that underlie these discrete actions. Specifically, the bucket movement motion recognition module analyzes the coordinated dynamics of the bucket, arm, and boom in egocentric videos using an LSTM network, which effectively captures sequential dependencies and subtle variations in motion continuity—critical factors for diagnosing the efficiency and smoothness of excavation tasks. In parallel, the swinging motion recognition module leverages dense optical flow to extract consistent background movement patterns, enabling robust detection of cab rotation and swing directions even in visually complex construction scenes. This two-stream design enables comprehensive motion understanding by combining temporal modeling (LSTM) and

spatial flow analysis (optical flow), allowing the framework to quantify how tasks are performed rather than merely what tasks are executed. Consequently, the framework establishes a direct linkage between motion-level dynamics and task-level productivity, advancing existing construction vision research from simple activity recognition toward diagnostic, data-driven productivity analysis.

The third contribution is that the proposed model offers a simple yet effective image-based approach to automate action and motion recognition. This suggests its potential for scope expansion to include other equipment and diverse construction tasks. In this sense, applying this approach to a loader's bucket, the most frequently used tool after an excavator, is feasible. This expandability could help develop an automated productivity analysis system encompassing various tasks on construction sites, which can contribute to digital-based construction management.

Consequently, the proposed framework advanced construction productivity analysis from simple time measurement to diagnostic intelligence. The framework generates productivity reports integrating action-level cycle analysis for overall productivity assessment, anomaly detection identifying underperforming cycles, and motion-level diagnosis with actionable feedback. This progression—from detection to diagnosis to recommendation—represents a fundamental shift in how productivity data can inform operational improvements and serves as a foundation for autonomous equipment training. Fig. 10 exemplifies a productivity report for Video #1, demonstrating how the framework translates recognition results into actionable insights.

- Part (A) Overview: This part provides an overview of the productivity analysis for the target earthmoving operation captured in an egocentric view. This part particularly includes basic information (e. g., excavator model, operator, duration, and site) about the earthmoving operation, the egocentric video with bounding boxes detected by the action module, and the time-series action and motion recognition results.

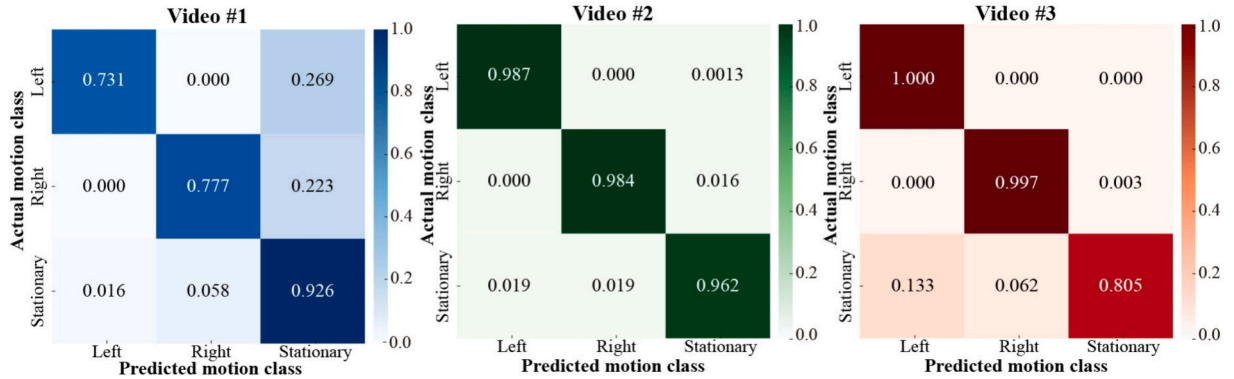


Fig. 8. Confusion matrices for swinging motion recognition in Videos #1, #2, and #3.

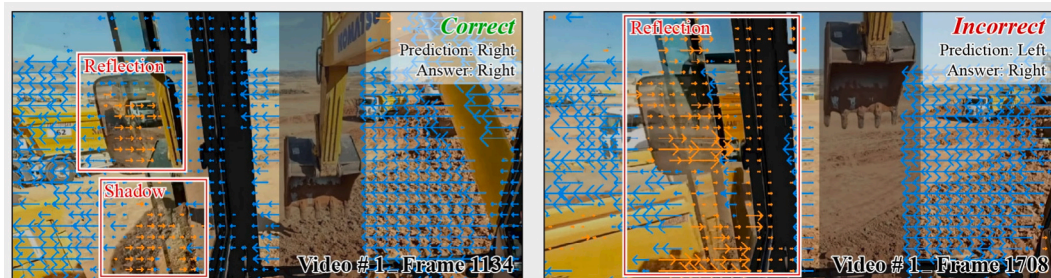


Fig. 9. Examples of correct and incorrect swinging motion recognition.

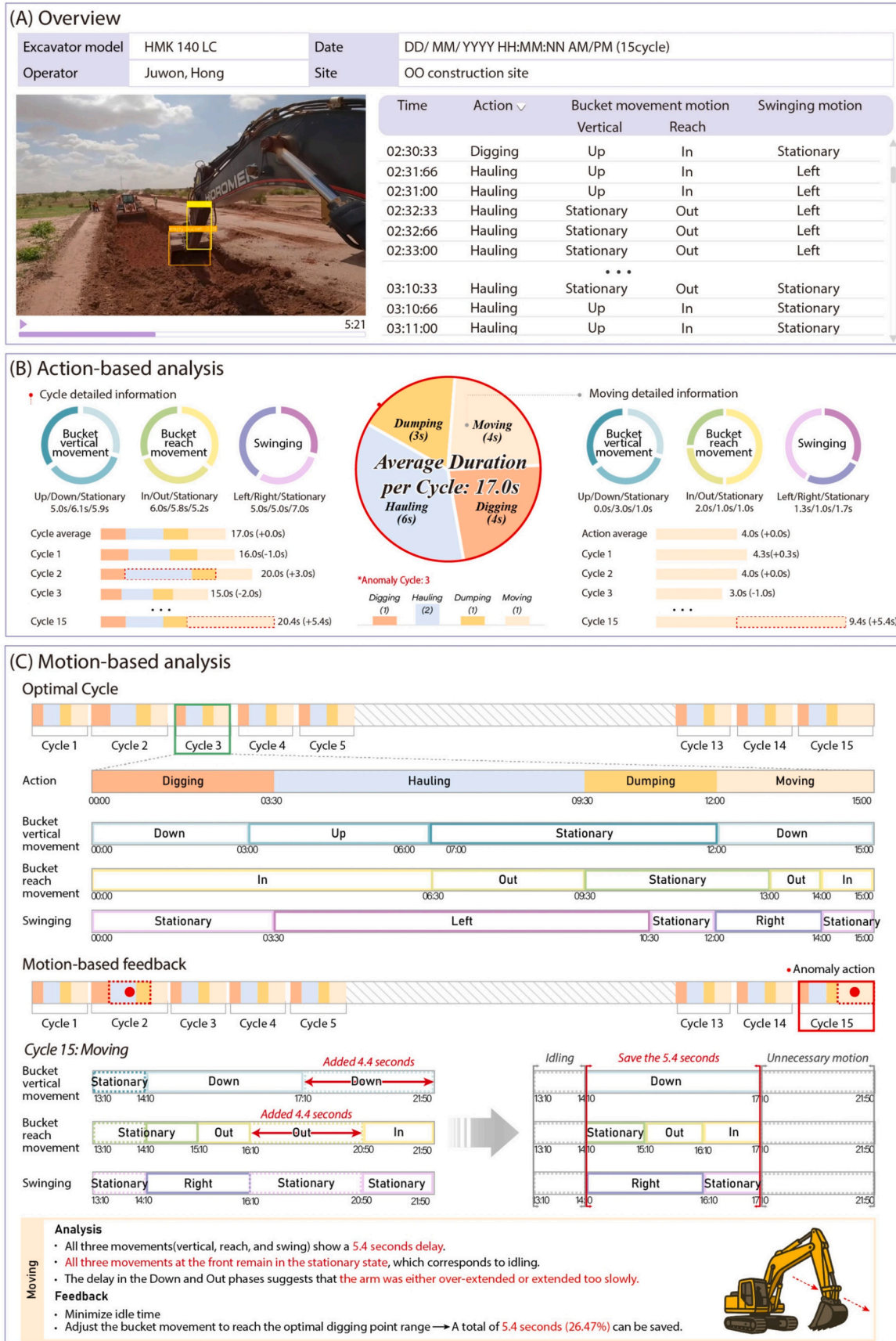


Fig. 10. Example of a productivity report for Video #1.

- Part (B) Action-based analysis: This part provides the results of an action-based analysis of various cycles that constitute earthmoving operation. The productivity analysis includes the average duration per cycle, the average duration for each type of action within a cycle, as well as the individual durations for each cycle and action. Cycles and actions with longer-than-usual durations (i.e., anomaly cycles and actions) can be identified by comparing average durations with individual durations. Additionally, more information can be included to elaborate on the composition of each cycle and action, as well as the degree of motion execution.
- Part (C): Motion-based analysis: This part provides detailed motion-level analysis of anomaly actions and presents actionable feedback benchmarked against the optimal cycle with the shortest duration due to the absence of unnecessary motion. For example, in an anomaly action labeled as “moving”, all motion segments in a certain interval were identified as “stationary” (i.e., idle), and the durations of the bucket’s “down” and “out” movements were longer than those in the optimal cycle. This suggests that unnecessary pauses occurred during the task, and that the bucket’s vertical and reach movements toward the digging point were performed slowly or over-extended. Based on the analysis, motion-based feedback is provided to minimize idle time and adjust the motion so that a bucket can reach its optimal digging range, with the expected improvement effect. Ultimately, such productivity reports can be actively utilized as fundamental data for work improvement, productivity enhancement, and autonomous equipment operations.

6.2. Limitations and future research

This study presents various implications of the proposed model, with the validity tested via experimental validation, and its applicability verified in terms of precise productivity analysis and practical improvements at the motion level. However, this study still has some technical and functional limitations, requiring further investigation. First, the proposed model exhibited excellent overall recognition performance for action and motion across different scenarios. However, in some areas, technical limitations remain and require further improvement. For example, action recognition leaves room for improvement in distinguishing between digging and moving. Then, in motion recognition, recognition of swinging motion (Video #1) and bucket reach movement (Video #2), depending on image characteristics, can be improved. In this context, further research may consider enhancing action and motion recognition performance by introducing auxiliary technologies, such as depth map estimation, or through technology advancements, such as more tailored bounding box object detection with higher precision. In this study, separate models were developed for motion recognition due to the distinct visual characteristics of each motion type; however, this approach may pose challenges in model implementation and maintenance. Therefore, future research should focus on developing an integrated model capable of recognizing multiple motions simultaneously by learning the overall visual context of egocentric videos. Second, since the proposed model aimed to recognize actions and motions with a certain interval (i.e., window), it primarily focused on identifying action state and directional motion patterns rather than quantifying their physical characteristics, such as speed or displacement. Consequently, it lacked other functions such as estimating the amount of soil in a bucket or predicting motion speed, which may be useful for productivity analysis and automation. Therefore, to expand the functionality of the proposed model, sensor fusion combining sensor data from various sources, including light detection and ranging (LiDAR) or stereo cameras for soil volume estimation, and IMU sensors for velocity and acceleration measurement, are suggested for future research.

7. Conclusions

Computer vision-based action recognition technology has gained much attention as a foundational technology for productivity enhancement and automation in the construction industry, extending beyond task-level productivity analysis, to enable real-time status recognition and autonomous operation of construction equipment. While action recognition technology developed based on third-person video has achieved a certain level of performance, the high variability and complexity of the construction environment, including object occlusion and overlapping issues, present significant challenges in systems for reliable recognition at construction sites. Therefore, to complement the inherent limitations of the third-person point of view and achieve more precise action recognition, this paper proposed a model capable of automatically recognizing actions as well as the detailed motions of the equipment performing the action based on egocentric videos captured from the operator’s perspective. The proposed model, designed for excavator operations, comprises action and motion recognition modules.

- Action recognition module: This module analyzes bucket states (e.g., empty, loaded, and dumping) and related features in egocentric videos to recognize action. Specifically, it detects bucket states using computer vision (i.e., YOLOv11), then applies rule-based classification based on the labels and properties of the bounding boxes to recognize each action (e.g., digging, hauling, dumping, and moving).
- Motion recognition module: This module recognizes three motion types (i.e., bucket vertical movement, bucket reach movement, and swinging) by analyzing temporal variations in the visual features of the bucket, arm and background captured in the egocentric video. The first sub-module utilizes an LSTM-based model to recognize bucket vertical movement motion (e.g., up, down, and stationary) and reach movement motion (e.g., in, out, and stationary) by analyzing temporal variations of arm and bucket bounding box properties, along with bounding box and action labels from the action recognition module. The second sub-module analyzes the optical flow of the background using the Gunnar Farneback algorithm-based optical flow model to recognize swinging motion (e.g., left, right, and stationary).

Experiments using three egocentric videos with different features quantitatively evaluated the action and motion recognition performance of the developed model, and the model’s validity and applicability were verified via experimental validation. The experimental results also demonstrated that the proposed model achieved high, stable, and consistent performance in action and motion recognition. The F1 scores were 0.928 for action recognition, 0.920 for vertical movement motion, 0.891 for reach movement motion, and 0.908 for swinging motion. These findings can be applied to generate more practical and data-driven productivity reports and are thus expected to contribute to improving various downstream tasks based on action recognition, as well as implementing autonomous driving systems and improving productivity using motion-based feedback. Nevertheless, this study has both technical limitations in recognizing actions and motions solely from egocentric videos and functional limitations in productivity analysis, as the proposed model focuses only on detecting actions and motions. Future research should address these issues by incorporating depth-aware vision models and multimodal sensing to enhance recognition robustness and generalizability.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea Government (MSIT; Ministry of Science and ICT) (NRF-2021R1A3B1076769).

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